Solutions and teaching notes, chapter 5

Teacher use. The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $\frac{12}{40} = \frac{3}{10} = 30\,\%$. *Action:* Link the partial count to the reference count.
2. $0{,}7$. *Action:* Represent the sample space split into two events.
3. $A \cap B$. *Action:* Shade the intersection of two sets.
4. $A \cup B$. *Action:* Include the common outcomes in the union.
5. $0{,}1$. *Action:* Replace the decimals with fractions.
6. $\frac{3}{6} = \frac{1}{2}$. *Action:* Identify the equally-likely assumption.

Investigation

Intersection needed 12; cells 12,28,18,42. Replacing with 13,27,17,43 keeps the margins but $\frac{13}{100} \neq 0{,}4 \times 0{,}3$.

Direct applications

1. $\frac{5}{60} = \frac{1}{12}$ and $\frac{15}{20} = \frac{1}{4}$.
2. $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$.
3. No: $0{,}4 \times 0{,}5 = 0{,}20 \neq 0{,}18$.
4. $(\frac{2}{3})^4 = \frac{16}{81}$ then $1 - \frac{16}{81} = \frac{65}{81}$.
5. $\frac{675}{2000} = 0{,}3375$; this is an estimate, not the exact value.

Exercises: methods and answers

Exercise 5.1 - A table. Cells: 15 (A and B), 25 (A without B), 20 (B without A), 40 (neither A nor B). $P_A(B) = \frac{15}{40} = \frac{3}{8}$; $P_B(A) = \frac{15}{35} = \frac{3}{7}$.

Exercise 5.2 - Production. $P(D) = 0{,}25 \times 0{,}04 + 0{,}75 \times 0{,}02 = 0{,}025$. $P_D(A) = \frac{0{,}01}{0{,}025} = 0{,}4$. The denominator changes when conditioning on D.

Exercise 5.3 - Independence to be checked. $P(A) = P(B) = \frac{1}{2}$; intersection $\{4; 6\}$ with probability $\frac{1}{} \neq \frac{1}{}$: not independent. The fact that another choice of threshold gave independence does not automatically carry over.

Exercise 5.4 - Without replacement. Two reds: $(\frac{3}{5})(\frac{2}{4}) = \frac{3}{10}$. One of each: $(\frac{3}{5})(\frac{2}{4}) + (\frac{2}{5})(\frac{3}{4}) = \frac{3}{5}$. The order distinguishes two disjoint paths.

Exercise 5.5 - Three trials. One success: three paths, each $(\frac{1}{3})(\frac{2}{3})^2 = \frac{4}{27}$, total $\frac{4}{9}$. Two success: $3(\frac{1}{3})^2(\frac{2}{3}) = \frac{2}{9}$; three: $1\overline{27}$; total $7\overline{27}$.

Exercise 5.6 - Detector. $P(S) = 0{,}02(0{,}9) + 0{,}98(0{,}05) = 0{,}067$. $P_S(D) = \frac{0{,}018}{0{,}067} = \frac{18}{67} \approx 26{,}9\,\%$. The low initial rate explains the gap from 90%.

Exercise 5.7 - Sets and cardinality. $A \cap B = \{2\}$ (1) ; $A \cup B = \{1;2;4\}$ (3) ; $E \setminus A = \{3;4\}$ (2); product $\{(1;2),(1;4),(2;2),(2;4)\}$ (4). A pair is ordered.

Exercise 5.8 - Delayed review. $P(B) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$. $P_B(A) = \frac{\frac{1}{8}}{\frac{1}{2}} = \frac{1}{4}$. The numerical equality with $P_A(B)$ is specific to these data.

CQFD, exercise 67 p. 336, questions 1, 2 and 3. $P(\overline{D} \cap \overline{N}) = 0{,}9 \times 0{,}05 = 0{,}045$. The probability of a positive test is 0 ,$1 \times 0{,}97 + 0{,}045 = 0{,}142$. Among positive tests, the probability of not being doped is $\frac{0{,}045}{0{,}142} = \frac{45}{142} \approx 31{,}7\%$. Do not swap positive and negative tests.

CQFD, exercise 70 p. 336, questions a and b. $P(B) = \frac{1}{4}$, $P_A(B) = \frac{1}{2}$ and $P_{\overline{A}}(B)$ = 0. Thus $\frac{1}{4} \neq \frac{1}{2}$ + 0, then 0 $\neq 1 - \frac{1}{2}$. The law of total probability involves the weights $P(A)$ and $P(\overline{A})$; the complement applies to B when the same condition is kept.

MCQ: answers and reasons

MCQ 1.

1. B. Reference sample space A.

2. A. Product along the branches.

3. D. Definition.

4. C. Criterion without division.

5. B. Empty intersection, positive product.

MCQ 2.

1. C. Complement of three faces.

2. A. Product p^2.

3. D. Probabilities of a conditional partition.

4. B. Observed frequency and model probability differ.

5. C. 2 red balls remain out of 4.

True or false

1. False: 12 common objects out of 40 A and 30 B give $\frac{3}{10}$ and $\frac{2}{5}$.

2. True: $P(A \cap \overline{B}) = P(A) - P(A \cap B) = P(A)(1 - P(B))$.

3. False: without replacement, the contents of the urn change.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.